

Stat 5200, Module 3: Completely Randomized Designs and ANOVA

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Overview

Objectives and overview

- This material is based Chapter 3 on the textbook “A First Course in Design and Analysis of Experiments” (Oehlert, 2000).
- Chapter outcomes include:
 - TODO

Completely Randomized Designs

Completely Randomized Design

- Completely randomized design (CRD) is the simplest design for comparing treatments and is sometimes the preferred design.
- Setup:
 - There are g treatment groups.
 - N experimental units (EU)s available.
 - You want sample sizes of $n_1, n_2, \dots, n_g$, with $\sum_{i=1}^g n_i = N$.

Completely Randomized Design II

- **Idea:** All possible sub-group composition are equally as likely.

Performing a Completely Randomized Design

- Randomly choose n_1 EU's for treatment 1.
 - Randomly choose n_2 EU's for treatment 2.
 - ...
 - Randomly choose n_{g-1} EU's for treatment $g - 1$.
 - The remaining n_g EU's are assigned to treatment g .
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- On a computer, this can be done by permuting $1, 2, \dots, N$, and assigning the first n_1 numbers to treatment 1, the next n_2 to treatment 2, etc.

Advantages and Disadvantages

- Advantages
 - Simple to do; all you need is one random permutation.
 - Simple to explain and understand.
 - Simple to analyze.
 - No restrictions on the sample sizes (other designs may restrict n_i).
 - Easy with missing values.
- Disadvantages
 - Not always possible or cost-effective.
 - The accuracy of the experiment depends on variability among the EUs, not just the variability among treatment groups. Thus, it may require more EUs (\$\$\$\$).

Example: Acid Rain

Wood and Bormann (1974) studied the effect of acid rain on trees. One experiment used 240 six-week-old yellow birch seedlings. These seedlings were divided into five groups of 48 *at random*, and the seedlings within each group received an acid mist treatment 6 hours a week for 17 weeks. The five treatments differed by mist pH: 4.7, 4.0, 3.3, 3.0, and 2.3. Otherwise, the seedlings were treated identically. After the 17 weeks, the seedlings were weighed, and total plant (dry) weight was taken as response.

Questions:

- What type of design is this, and can you describe why?
- What are the EUs, and how many are there?
- What was the response?

Example: Acid Rain II

- What compromises may have been made in this case?

Like many experiments, the researchers necessarily have to narrow down an enormous question to a workable experiment using artificial acid rain under controlled conditions. For any experiment, a considerable amount of non-statistical background work goes into the planning even the simplest experiment.

Models and Parameters

Introduction and definitions

- All formal statistical analysis is based on probability models.
- *EX*: The number of heads in ten tosses of a fair coin would be $\text{Binomial}(10, 0.5)$. Here, the model depends on 2 parameters: $N = 10$ the number of trials, $p = 0.5$ is the probability of heads.
- In an experiment, we may posit several different models for the data, all with *unknown* parameters. One of our objectives is often to describe which model is the best description of the data, and make *inference* about parameters.

Definition: Inference

Using data to learn properties of a probability distribution. In our case, using data to estimate parameters of a model.

Introduction and definitions II

- The models we will consider for experimental design will have two basic parts:
 - A description of the expected value, or mean, of the data. Sometimes called “model for the means”.
 - A description of how data vary around the treatment means. Sometimes called ‘model for the errors’.
- For the errors, we will generally assume that they are independent for different data values, have mean zero, and all have a common variance σ^2 .
- To perform formal tests and obtain confidence intervals, we will further assume a distribution of the errors. Generally, we assume that the errors follow a **Gaussian** (Normal) distribution.

Introduction and definitions III

- For a CRD, we start with two basic models.
 - Recall: g treatment groups, N EUs.
1. **The Full Model** (separate group means)

Introduction and definitions IV

2. The Reduced Model (single mean model)

Defining overall mean

- It's sometimes useful to express the group means μ_i as:

$$\mu_i = \mu^* + \alpha_i.$$

In this case, we call μ^* the *overall mean*, and the α_i 's are the *i*th *treatment effects*.

- This introduces an extra parameter ($g+1$ total), which we need to address. Generally, we do this by defining μ^* and require that $\alpha_i = \mu_i - \mu^*$.

Defining overall mean II

- One classic choice is to define μ^* as the mean of the treatment means:

Defining overall mean III

- An alternative choice (that can sometimes make hand-work simpler) instead describes μ^* as the weighted average:

Defining overall mean IV

- Another common choice in software is to set $\mu^* = \mu_j$, for some $j \in 1 : g$. Many software just use the last mean: $\mu^* = \mu_g$.

Model Recap

However you choose to define μ^* and α_i 's, we end up with the following models for the data:

- *The full model:*

$$y_{ij} = \mu^* + \alpha_i + \epsilon_{ij}$$

- *The reduced model:*

$$y_{ij} = \mu + \epsilon_{ij}$$

Parameter Estimation

Estimating parameters

- We want to use data to estimate μ_i 's and the variance σ^2 .
From these, we'll compute μ^* and α_i 's.
 - We will focus on computing **unbiased** estimates of these parameters.
 - Estimate for μ_i :
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- The estimate for μ^* is fixed, so it depends on how we defined it.

Estimating parameters II

- Unweighted average
- Weighted average
- Reference Group

Estimating parameters III

- The only parameter left to estimate is σ^2 :

Estimating parameters IV

- Above, we have described **point estimates**. In some sense, these are our “best guess” as to the value of a particular parameter.
- We also want some measure of uncertainty of this estimate. The standard frequentist way of doing this is a **confidence interval**.
- Confidence intervals for α are not very useful, because it depends on how we define them. Confidence intervals for group means are typically defined as:

Comparing Models: ANOVA

Analysis of Variance

- In our analysis of a CRD, our models focus on the *mean responses* of the treatment groups.
- A basic question we might want answered: Are all the means the same, or are some different?
- We can express this same question mathematically:

Analysis of Variance II

- The basic idea for many statistical tests (including ANOVA) follows *Occam's Razor*: we use the simplest model that is consistent with the data; we will only use a more complicated model if the data indicate that the more complicated model is needed.
- How do we know if a more complex model is needed? We compare **residuals**.

Analysis of Variance III

- The lower the SSR, the more of the variance in the data is explained by the model.
- Importantly, the SSR for the reduced (single mean) model can *never* be lower than the separate model, and in practice, is nearly always larger.
- What we need now is a way to consistently decide when the separate means model is sufficiently “better” than the single means model.
- To do so, we will set up a statistical hypothesis test:

ANOVA mechanics

- ANOVA works by partitioning the total variance in the data into parts that mimic the model.
- For instance, in the separate mean model, the model says the data differ from the grand mean by a random error and a treatment effect:

ANOVA will have a similar decomposition: one part dealing with group means, one part with deviations from those means:

- Recall from Eq. ??, we have:

$$y_{ij} - \bar{y}_{\bullet\bullet} = \hat{\alpha}_i + r_{ij},$$

ANOVA mechanics III

- Finally, we can connect this back to the residuals.
- The Sum of Squared Residuals (SSR) depended on our model:

$$SSR_{\text{single means}} = \sum_{i=1}^g \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{\bullet\bullet})^2 = SS_T.$$

$$SSR_{\text{separate means}} = \sum_{i=1}^g \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{i\bullet})^2 = SS_E$$

Calibrating Effect Size

- We have established:
 - We will compare single mean against separate mean by comparing their SSRs.
 - Comparing SSR can be done by decomposing total error into two parts: $SS_T = SS_{T_{rt}} + SS_E$, and if the separate model is “much better” than the $SS_{T_{rt}}$ will be large.
- **How large is large enough?**

Calibrating Effect Size II

- The answer also lies in the decomposition of sum of squares.

ANOVA variable recap

- Sum of Square Residuals:

$$SSR = \sum_{i=1}^g \sum_{j=1}^{n_i} (y_{ij} - \hat{y}_{ij})^2$$

- Treatment Sum of Squares:

$$SS_{Trt} = \sum_{i=1}^g \sum_{j=1}^{n_i} (\bar{y}_{i\bullet} - \bar{y}_{\bullet\bullet})^2 = \sum_{i=1}^g n_i \hat{\alpha}_i^2.$$

- Error Sum of Squares:

$$SS_E = \sum_{i=1}^g \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{i\bullet})^2$$

ANOVA variable recap II

- Total Sum of Squares:

$$SS_T = SS_{Trt} + S_E = \sum_{i=1}^g \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{\bullet\bullet})^2$$

- F-statistic:

$$\frac{SS_{Trt}/(g-1)}{SS_E/(N-g)} \sim F_{g-1, N-g}.$$

Contrasts

Introduction to Contrasts

- An ANOVA can give us a good overview as to whether all the treatment groups have the same mean response.
- However, an ANOVA by itself does not tell which treatments are different or in what ways they differ.
- One method to examine treatment effects is called a **contrast**.
- A contrast is a formula which compares two or more treatment means or effects in a way that does not depend on the overall level μ .

Formal Definition

- A contrast is a linear combination of treatment means μ_i 's:

$$W(\mu_i) = \sum_i w_i \mu_i, \quad \sum_i w_i = 0$$

- The condition $\sum_i w_i = 0$ guarantees that $W(\mu_i)$ does not depend on μ .
- Often the weights w_i themselves are referred to as a contrast.
- An observed contrast or estimated contrast is defined as:

$$W(\bar{y}_{i\bullet}) = \sum_i w_i \bar{y}_{i\bullet} = \sum_i w_i \hat{\alpha}_i = W(\{\hat{\alpha}_i\})$$

Common Contrasts

Types of Common Contrasts

Pairwise Contrasts

- It is often of interest to compare all pairs of groups.
- For a total of g treatment groups, there are $\frac{1}{2}g(g - 1)$ different pairwise contrasts.

Types of Common Contrasts II

Comparison with a Control

- Let's suppose treatment 1 is a control.
- An obvious idea is to compare the mean or effect of the control with the average mean or effect of all non-controls.
- Individual pairwise comparisons with the control would probably be of interest too.
- Notice that multiplying the coefficients of a contrast by a common factor (usually to produce integer coefficients) will result in an equivalent contrast.

Types of Common Contrasts III

Factorial Treatments

- Suppose there are two treatment factors, A and B, each at two levels.
- There will be 4 treatments with means $\mu_{11}, \mu_{12}, \mu_{21}, \mu_{22}$.
- Natural contrasts in such experiments include:
 - Average of row 1 vs average of row 2.
 - Average of column 1 vs average of column 2.
 - Difference between effects of A for the two levels of B.
- When this contrast is not zero, it means the effect of A depends on the level of B, or equivalently, the effect of B depends on the level of A.
- When such a situation occurs, we say there is interaction between factors A and B, and we call it an interaction contrast.

Linear Contrasts

- Suppose the treatments are determined by a quantitative variable X with levels $x_1, x_2, \dots, x_g$.
- Then, if you fit a regression line, the least squares estimate of the slope is:

$$b = \frac{\sum_i n_i (x_i - \bar{x}) \bar{y}_i}{\sum_i n_i (x_i - \bar{x})^2}$$

- This is a contrast with weights proportional to $(x_i - \bar{x})$.
- This is called a linear contrast.
- It will be large when there is a high degree of linear dependence of the means on X .

Variance and Confidence Intervals

Inference on Contrasts

- Under the null hypothesis and normality assumption, we can use Student's t with $df = N - g$ in tests or confidence intervals.
- Under the null hypothesis, $W(\mu_i) = \sum_i w_i \mu_i = 0$.
- The standardized estimated contrast follows a student-t distribution with $df = N - g$:

$$t = \frac{\sum_i w_i \bar{y}_{i\bullet}}{s_p \sqrt{\sum_i w_i^2 / n_i}}$$

Inference on Contrasts II

- A $1 - \alpha$ confidence interval for a contrast $W(\mu_i)$ has the usual form:

estimate $\pm$ critical value $\times$ standard error

Orthogonality

Orthogonal Contrasts

- Two contrasts $W^{(1)}$ and $W^{(2)}$ are orthogonal if:

$$\sum_{i=1}^g \frac{w_i^{(1)} w_i^{(2)}}{n_i} = 0$$

- When the sample sizes are equal, this simplifies to:

$$\sum_{i=1}^g w_i^{(1)} w_i^{(2)} = 0$$

- When the ANOVA assumptions are satisfied, two orthogonal contrasts are independent.
- With independent errors (but not normality), orthogonal contrasts are uncorrelated.

Orthogonal Contrasts II

- **Example:** With a factorial design, there are $g = 4$ treatments defined by all combinations of two factors A and B each at two levels.
- An important family of contrasts for this case are defined by:
 - $W^{(1)} = \{\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}\}$ or $\{1, 1, -1, -1\}$.
 - $W^{(2)} = \{\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}\}$ or $\{1, -1, 1, -1\}$.
 - $W^{(3)} = \{1, -1, -1, 1\}$.

Polynomial Contrasts

Polynomial Contrasts

- Polynomial contrasts are particularly useful when the treatments are determined by levels of a quantitative variable X .
- We looked at a linear contrast, whose coefficients are proportional to $w_i = x_i - \bar{x}$.
- Similarly, there are contrasts that focus on the quadratic, cubic and higher order terms.
- When treatments are equally spaced, there are tables of polynomial contrast coefficients (Table D6, pp. 630).
- But these don't help when the treatments are not equally spaced.

Polynomial Contrasts II

- In either case, the SAS function `orpol` can compute the coefficients of a polynomial contrast.
- Polynomial contrasts are all mutually orthogonal.
- **Note:** Polynomial contrasts should be tested from higher order to lower order and stopped at the first significant order.

References and Acknowledgements

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References and Acknowledgements II

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